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TWELVE TIPS



Twelve tips for interpreting abdominal CT scans

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ABSTRACT

Background: Abdominal computerised tomography (CT) scans are a crucial tool in the diagnosis and management of the acute abdomen. Currently, medical students are not widely and extensively trained in the interpretation of abdominal scans.

Aim: We aim to provide advice about interpreting abdominal CT scans.

Methods: We used the critical reflection of our experiences, both in clinical practice and in teaching, alongside advice from the literature to develop these tips.

Results: Twelve tips following the '4As, 3Bs, 2Cs and 1D' approach are presented to assist doctors and medical students with interpreting abdominal CT scans.

Conclusion: The early identification of pathology on CT scans has been demonstrated to improve patient outcomes in certain cases, while a formal radiologist's report is awaited. Following a systematic approach, such as the one we presented here, may aid trainees in looking at abdominal CT scans.

KEYWORDS

CT abdomen; interpreting abdominal CT scans; systematic approach to CT scans; abdominal computerised tomography interpretation

Introduction

Abdominal computerised tomography (CT) scans are a core component of the modern surgical assessment of the acute abdomen, which is offered to a significant proportion of patients (Stoker et al. 2009; Paolantonio et al. 2016).

In spite of that, evidence suggests that on average only 5% of total teaching time in medical schools in the UK is dedicated to radiology (Heptonstall et al. 2016), with students reporting low confidence in interpreting abdominal imaging (Jacob et al. 2016) and performing poorly at the recognition of life-threatening CT images (Nguyen et al. 2017). At the same time, interpreting abdominal CTs is considered to be an important skill by residency program directors in the US (Kondo and Swerdlow 2013).

We developed a basic systematic approach to viewing abdominal and pelvic CT scans, which can be taught to junior members of the medical profession. It follows a simple '4As, 3Bs, 2Cs and 1D' approach. It is not, however, intended as a comprehensive methodology for interpreting abdominal/pelvic CT scans, and formal reports should be sought where possible.

Tip 1

Perform the pre-interpretation checks

It is vital to make sure that pre-interpretation checks are made. These include: name and date of birth of the patient, date of scan, type of scan and images—are there arterial and venous phase scans or pre- and post-contrast images and if so, are you viewing the right one? Finally, does the scan allow you to see what you want to see, for

example, is it a contrast scan for renal colic? If so, it may not be helpful.

The four 'A's

Tip 2

The 1st A: air. Assess for (1) pneumoperitoneum, (2) intramural bowel gas and (3) fluid-gas level within collections

Firstly, the CT scan needs to be assessed for pneumoperitoneum (free gas in the abdominal cavity). It is a sign of perforation of a hollow viscus and prompt identification of this sign is crucial (Tanner et al. 2018). In order to maximise the ability to see pneumoperitoneum, consider changing the image window to 'lungs', which will make air appear black in contrast to the relatively bright abdominal viscera. Look at all the images to ensure that small localised perforations are not missed.

Some air within the abdominal cavity is not pathological, such as gas within the bowel lumen and free air after laparoscopic surgery. It has previously been reported that 23% of post-laparoscopy CT scans have pneumoperitoneum visible for up to 3 weeks post-operatively with only a small percentage of these patients requiring any intervention for this finding (Chapman et al. 2015).

Secondly, check for gas contained within the bowel wall. *Pneumatosis intestinalis* (intramural bowel gas) appears as gas bubbles within the bowel wall, indicating a disruption in the integrity of bowel mucosa which leads to increased permeability. It may indicate bowel ischaemia, necrosis or, less commonly infection and others (Treyaud et al. 2017). Its management can be challenging, but

emergency surgery is often required (Tahiri et al. 2015; Ross et al. 2018).

Thirdly, one should also check for air within a fluid collection, in the form of an air-fluid level or 'bubbles', as this can be a sign of gas-forming organisms within the fluid, indicating an active infection.

Tip 3

The 2nd A: aorta. Assess for (1) aneurysms and (2) atherosclerosis

It is important to follow the aorta from the aortic valve (if visible on the scan) down to the common iliac bifurcation and then along each of the iliac/femoral vessels until the inferior most image of the scan. Observe for signs of an abdominal aortic aneurysm and any associated leakage of blood, as well as the 'tennis ball sign' of aortic dissection (Lal et al. 2017). Aneurysms can occur anywhere along the vessel and may involve the aorta or any of its branches, to the common iliac arteries and beyond.

In addition to aneurysms, identification of the main branches of the aorta allows for other potential causes of abdominal pain to be seen, such as mesenteric ischaemia and infarction. In non-contrast images atherosclerotic calcification of these vessels, often at their origin, can raise the suspicion of potential vascular compromise (Wang et al. 2018). In the presence of contrast look for transition areas within the vascular tree, indicating the loss of blood flow.

Tip 4

The 3rd A: ascites. Check if there is any fluid in the abdomen

Ask yourself two questions: (1) is the fluid ascites or blood?; (2) if it is ascites, is it exudative or transudative? Identifying the presence and the likely origin of fluid in the abdomen can give clinicians useful clues to the underlying pathology and the necessary management.

Firstly, it is important to try to decide whether the fluid is ascites or blood. A large volume of intraabdominal fluid without a compensatory haemodynamic response (tachycardia $\pm$ hypotension) generally indicates ascites, rather than blood. If the volume is small, it can be difficult to distinguish ascites from hemoperitoneum on the scan alone, but clinical assessment may give further clues.

Secondly, it may be beneficial to establish if the ascites is transudative or exudative. This is also quite difficult based on the scan alone, however one way is to note if the fluid is loculated or not. The presence of septa within the fluid may indicate an exudative cause, but fluid can also become loculated through its confinement by adhesions or malignancy (Rudralingam et al. 2017) and the lack of septations does not rule out an infective element.

Tip 5

The 4th A: appendix. Assess this common site of pathology

This is an important step due to the relatively high incidence of acute appendicitis paired with the ability of this condition to mimic a large variety of other conditions (Bhangu et al. 2015).

The appendix, as an anatomical variant, can be found in many locations throughout the abdominal cavity including the pelvis, posterior to the urinary bladder, and in the left upper quadrant (Barlow et al. 2013). Therefore, it is often easier to follow the colon proximally from the rectum until the ileo-caecal junction, at which point the appendix can be identified as a blind-ended appendage of the caecum. Signs of inflammation include peri-appendiceal fat stranding, extraluminal air, peri-appendiceal fluid or appendix abscess (Kim et al. 2018).

The three 'B's

Tip 6

The 1st B: bowel. Assess the bowel for pathology

When looking at the bowel, consider the following: (1) look at the whole bowel and (2) focus on the area which indicates pathology from your clinical assessment.

The bowel needs to be followed from the pylorus to the rectum in a systematic manner. It is often possible to use clinical signs to pinpoint the likely point of concern, but the presentation can be misleading, for example in cases of pelvic inflammatory disease due to pus collection in the dependent areas of the peritoneal cavity secondary to peptic or duodenal ulcer perforation (Su et al. 2005).

It is important to be able to clinically identify the signs of perforation, obstruction, bowel infarction and large malignancy encroaching on the bowel lumen, amongst others. These common conditions can all be part of the acute abdomen and identifying them prior to the full radiologist's report allows for basic interventions, such as nasogastric tube and intravenous fluids in bowel obstruction, to be instigated to mitigate the potential damage.

Tip 7

The 2nd B: bladder. Assess the kidneys, bladder and collecting system

Diseases of the genitourinary tract are among the top 10 primary inpatient diagnoses in the UK (NHS Digital 2019b) and therefore the urological tract is one of the areas of the abdominal CT scan that needs to be carefully considered.

Initially, identify the kidneys and note any asymmetry in size or location and any anatomical variations. Look for the bladder and any signs of obvious pathology such as cysts, tumours or gross dilatation. Then identify the collecting system of one of the kidneys and follow the ureter closely from the renal pelvis to the bladder. Repeat this for the opposite side. Look for any obvious dilatation and/or asymmetry in size or location of the kidneys and ureters, potentially indicating polycystic kidneys, a tumour or obstructive uropathy. Also look for signs of calculi, such as bright white signal in the lumen of the renal pelvis, ureter or bladder. If stone disease is suspected clinically and there is contrast present within the lumen of the ureter it can make the test difficult to interpret, unless there are signs of absolute blockage with no contrast after a certain point.

Tip 8

The 3rd B: biliary tree. Look for pathology of the (1) common bile duct and hepatic duct and (2) gall bladder

The biliary tree is another common cause of patients presenting with an acute abdomen (NHS Digital 2019a). Common diagnoses include choledocholithiasis, obstruction from external compression such as in pancreatic malignancy or sphincter of Oddi stricture, or biliary sepsis, usually secondary to cholecystitis and ascending cholangitis.

The entire biliary tree should be identified, starting from the sphincter of Oddi and working proximally to notice any common bile duct or hepatic duct dilatation or collapse, as well as identifying the gall bladder and any signs of perforation—in the form of pericholecystic fluid—or inflammation, in the form of fat stranding around the gall bladder (Morris et al. 2007).

The two 'C's**Tip 9**

The 1st C: cancer. Check for any sign of malignancy

We recommend the following: (1) identify gross abnormalities within abdominal and pelvic organs, as well as abdominal wall; (2) consider whether it is likely to be a malignant (primary or metastatic) or non-malignant lesion.

Examples of gross abnormalities that may be identified include malignancies within the liver, both primary and metastases; malignancies of the kidneys, bowel, ovaries or stomach, as well as tumours in the abdominal wall, to name a few. Metastatic masses can grow rapidly, have a similar radiological appearance to their primary site lesion, and should especially be considered in the differential if the patient has a history of malignancy and/or has other sites of metastases (Li et al. 2019). Non-radiology trainees, however, are not expected to be competent at identifying the nuances of small malignancies.

In the acute admission, cancers are usually diagnosed based on their sequelae, for example complete bowel obstruction in colorectal malignancy (Guimaraes et al. 2014), or as a clinical sign, such as obstructive jaundice in a patient with liver malignancy (Lai and Lau 2006). However, the diagnosis of malignancy can be incidental, such as the finding of a markedly enlarged kidney relative to the contralateral organ in the case of renal cell carcinoma.

Early identification of neoplasms and their sequelae may aid decision making, but usually warrants conservative treatment until a full radiologist report is available. An exception might be cases of faecal peritonitis, where an emergency laparotomy may be indicated, following appropriate escalation to a senior and a direct discussion with a radiologist (Ross et al. 2018).

Remember that the differential for masses in the abdominal wall includes benign tumours, such as lipomas, as well as tumour-like lesions, such as hernias and endometriosis (Li et al. 2019).

Tip 10

The 2nd C: cutaneous/soft tissue. Examine the images for pathology within the skin and soft tissues of the abdominal wall

Once the internal aspect of the peritoneal cavity has been interrogated, examining the images for problems within the skin and soft tissues of the abdominal wall may elucidate the cause of the presenting complaint or a second abnormality, either related or not.

Examples of causes of presentations resembling the acute abdomen from these structures include rectus sheath haematomas, abdominal wall cellulitis, or an injury that did not penetrate the parietal peritoneum.

The one 'D'**Tip 11**

The D: doctors' interventions. Check (1) drains, (2) signs of previous surgery, (3) complications of recent interventions

Look for any signs of drains (including biliary, ascitic, post-operative, etc.) within the abdominal cavity. These can be signs of active pathology and may need to be investigated further as iatrogenic causes of a patient's presentation—e.g. is the drain blocked? Is it in the wrong location? Surgical clips or intravascular prostheses can be signs of an underlying disease and may modify the differential diagnosis.

In post-operative patients, it is particularly important to consider iatrogenic causes for their presentation. For example, surgical procedures performed around the pelvic region and retroperitoneal abdominal space can lead to urinary tract injuries (Esparaz et al. 2015). Post-pancreatic surgery complications include bile leak, bile tract obstruction and infection (Angileri et al. 2019).

Bringing it all together**Tip 12**

Summarise the findings

Once all 10 aspects of the CT scan have been systematically reviewed, the next step is to corroborate the findings of the CT scan with the clinical history and examination, and any other investigations.

Usually the interpretation of CT images for patients with an acute abdomen by a non-radiologist will allow for initial management to be started. Should imaging point to a pathology that may require surgical intervention, appropriate senior escalation and discussion with a radiologist to confirm the diagnosis is necessary, especially if there is any ambiguity.

Conclusions

We presented a simple structure for looking at abdominal and pelvic CTs, which can aid junior members of the medical profession. The importance of ensuring that junior trainees receive good-quality radiology training is widely recognised (Jacob et al. 2016). It has been demonstrated that the cautious use of CT findings prior to a formal

radiology report by staff in the emergency department improves outcomes by early recognition and treatment (Bagheri-Hariri et al. 2017).

Disclosure statement

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the article.

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